CIVICEYE

Smart Village Complaint Management System

Complete master prompt for Google AI Studio

How to use this document

Copy the complete master prompt below and paste it into Google AI Studio. Attach the CIVICEYE 20-screen reference image as the visual guide. If the full build exceeds one generation, ask the builder to implement it in stages while preserving the same project structure and all previously completed features.

MASTER PROMPT — COPY FROM HERE

PROJECT GOAL

Build a complete, production-ready application named CIVICEYE — Smart Village Complaint Management System. It must help village residents report local public-service problems and help authorised administrators manage complaints from submission to resolution. Use the attached 20-screen CIVICEYE reference image as the main visual inspiration. Do not build a generic social network, e-commerce app or broad city-services portal. Keep the scope focused on village municipal services, electricity works and citizen-submitted other local problems.

TECHNOLOGY AND DELIVERY

Build a responsive web application that works smoothly on Android phones, tablets and desktop browsers and can be installed as a Progressive Web App (PWA) where supported. Use React, TypeScript and a clean component-based architecture. Use Firebase Authentication for sign-in, Cloud Firestore for application data, Firebase Storage for complaint photographs and Firebase Cloud Messaging (FCM) for push notifications where configured. Use Google Maps Platform for map display, geocoding and location selection. Keep API keys and secrets out of client code; use environment variables and document required configuration. If a service key is unavailable, show a clear setup state rather than pretending the integration works.

VISUAL DESIGN AND NAVIGATION

Follow the reference design: professional blue-and-white palette, clean typography, rounded cards and buttons, subtle shadows, clear icons, accessible contrast and consistent spacing. Make the interface simple for villagers with limited technical experience. Use a mobile-first layout, bottom navigation for citizen screens and clear admin navigation/sidebar on larger screens. Include loading, empty, success, validation and error states. Provide back navigation, confirmation dialogs for important actions and useful form labels. Maintain CIVICEYE branding throughout.

USER ROLES AND AUTHENTICATION

Provide two primary roles: Citizen and Admin/Officer. Show a role-selection screen, but never allow a user to grant themselves admin privileges by selecting a role. Citizen registration should collect name, mobile number, optional email, village/ward or locality and password or phone OTP as configured. Implement secure login, logout, password reset and session persistence. Admin accounts and roles must be created or assigned only through a trusted, secure process (such as Firebase custom claims or an authorised admin provisioning workflow). After login, route users to the correct dashboard based on their verified role. Protect every admin route and database operation with role-based authorization.

CITIZEN MODULE

Create these screens and working flows: splash/welcome screen; role selection; citizen registration; login; OTP verification when phone authentication is configured; citizen home dashboard; report-a-problem category selection; complaint details form; photo capture/upload; GPS/map location selection; submission confirmation with unique complaint ID; My Complaints with All, Submitted, Verified, In Progress and Resolved filters; complaint details and status history; nearby reported issues map; notifications; emergency information/contact screen; profile view/edit; help and logout. The dashboard should provide clear shortcuts for Report an Issue, My Complaints, Nearby Issues and Emergency Information, plus recent complaint updates.

COMPLAINT CATEGORIES — STRICT PROJECT SCOPE

Provide these main categories only: (1) Garbage Collection; (2) Road Damage and Repairs; (3) Street Light Problems; (4) Water Supply Problems; (5) Drainage and Sewage Problems; (6) Electricity Services / Electrical Works; and (7) Other Services. Include relevant optional subcategories: garbage not collected/overflowing bins; potholes/damaged roads; street light not working/damaged poles; water interruption/leakage/contamination; blocked drains/sewage overflow; power outage/damaged electric poles/faulty transformers/damaged service lines/exposed wires. The Other Services option must let citizens enter a custom problem title, explain the issue, optionally identify the relevant local service/department, attach photos and provide the location. Do not force an Other Services complaint into a predefined category. Save its custom title and description clearly.

SUBMIT A COMPLAINT

Every complaint form must collect category, problem description, village/locality or address, exact map pin/GPS coordinates where available, and at least one relevant photograph when possible. Allow camera capture or gallery upload, preview and remove/replace images, validate file type and size, and show upload progress. Allow a citizen to correct the map pin or enter a location manually if GPS is unavailable or permission is denied. Generate a readable, unique complaint reference ID and server timestamp. Prevent accidental duplicate submissions caused by repeated taps. Before submission, show a review of the complaint details. After successful save, show the reference ID and a link to view the complaint. Never display success unless the database write and required upload have actually succeeded.

COMPLAINT STATUS AND TRACKING

Use a controlled status lifecycle: Submitted → Verified → In Progress → Resolved. Admins may reject or mark a complaint as Duplicate only with a mandatory reason; show these outcomes transparently to the citizen. Record every status change with actor, timestamp, previous status, new status and optional note. Citizens can view the current status, priority, assigned department/worker where appropriate, submission date, updates and resolution comments. Status changes must update the citizen view in real time using Firestore listeners and create an in-app notification. Do not let citizens change official status values.

ADMIN / OFFICER MODULE

Create a secure admin login and dashboard showing total complaints, new/submitted, verified, in progress and resolved counts. Provide a searchable, filterable complaint list by category, status, priority, date and locality. Admins can open complaint details, view photos and map location, verify or reject with a reason, detect/mark duplicates, assign priority (Low, Medium, High, Urgent), assign a responsible department or worker, update status, add internal notes and citizen-visible comments, and mark a complaint Resolved with a resolution note and optional after-photo. Include confirmation before final resolution. Provide a complaint activity timeline and basic reports showing category totals, status totals and complaint trends. Allow authorised admins to publish village-wide announcements and urgent public-service notices. Admins must not be able to access citizen passwords or authentication secrets.

ELECTRICITY AND EMERGENCY SAFETY

Treat electrical complaints as potentially hazardous. Show a prominent safety message for exposed wires, fallen electric poles, sparks, fire or damaged transformers: do not touch or approach the hazard; keep others away and contact the appropriate electricity emergency service immediately. Provide configurable local emergency contact numbers managed by an authorised admin. Do not invent official helpline numbers. CIVICEYE is a reporting and tracking platform, not an emergency dispatch service, and must clearly say that submitting an app complaint does not replace calling emergency services for immediate danger.

MAP, LOCATION AND NEARBY ISSUES

Show complaint pins on a map using stored coordinates. Display only appropriate public complaint information on the nearby map; never expose a citizen's private phone number, email or personal profile. Allow map/list view, category filters and complaint detail previews. Ask permission before accessing device location. Provide manual location entry when permission is unavailable. Do not fabricate coordinates or claim a location was captured when it was not.

NOTIFICATIONS AND ANNOUNCEMENTS

Create an in-app notification centre for complaint submission, verification, assignment where citizen-visible, status changes, resolution and admin announcements. Show read/unread state and allow Mark All as Read. Use push notifications only when Firebase Cloud Messaging is correctly configured and permission is granted. Store notification records in Firestore so users can still see them in-app. Announcements should include title, message, category/severity, publish date and optional expiry date.

DATABASE AND SECURITY

Design Firestore collections/documents for users, complaints, complaintStatusHistory, workers or department assignments, notifications and announcements. Each complaint should include complaintId, citizenId, category, customCategoryTitle when applicable, subcategory, description, image URLs, address/locality, latitude, longitude, status, priority, assignedDepartment, assignedWorkerId if used, createdAt, updatedAt and resolution details. Use server timestamps where possible. Add Firestore and Storage security rules: citizens can read/update their own permitted profile fields and read their own complaints; citizens can create complaints only for themselves and cannot edit official status, priority or admin fields; admins can manage complaints and announcements according to verified roles; uploaded files must be authenticated and validated. Minimise personal data and never expose private citizen data publicly.

QUALITY, TESTING AND COMPLETION

Implement real navigation, functional buttons, form validation, authentication, database CRUD, image uploads, maps, filtering, search and status tracking. Do not use hard-coded demo data as if it were live data. If sample data is included, label it clearly as demo data and provide a way to remove it. Handle offline/slow network states gracefully and prevent data loss during form submission. Test the complete citizen journey from registration to complaint resolution and the complete admin journey from login to resolution. Fix console errors, TypeScript errors and broken routes. Provide a README with setup instructions, Firebase project configuration, required environment variables, Google Maps setup, Firebase security rules, notification setup and deployment steps. Build all screens and connect them into one complete application, not merely a static UI prototype.

IMPLEMENTATION INSTRUCTIONS

First inspect the reference image and plan the application structure. Then implement the shared design system, authentication and role protection, citizen module, admin module, Firebase data layer, maps, notifications and testing. Keep code modular and maintainable. If the entire project cannot be completed in one generation, continue in clearly defined stages without replacing working code or removing earlier features. At the end, list what is implemented, what external credentials/configuration the owner must supply, and any feature that remains incomplete. Never claim an unconfigured integration is operational.

Build acceptance checklist

- Citizen and admin accounts have separate, secure permissions.
- Only the village-focused categories are shown, including Electricity Services and flexible Other Services.
- Citizens can submit complaints with description, photo and location, and receive a unique reference ID.
- Citizens can track status history and receive notifications.
- Admins can verify, prioritise, assign, update and resolve complaints.
- Map views do not expose private citizen information.
- Electrical hazard warnings and configurable emergency contacts are present.
- Firebase rules protect citizen and admin data.
- All visible controls work; there are no fake success messages or dead-end screens.
- Setup, configuration, testing and deployment instructions are documented.

END OF MASTER PROMPT